

Aim:- Implementation of Java & syntax of Java Program

Introduction to Java :- Java is a high-level, object-oriented & platform-independent programming language. It follows the principle "Write Once, Run Anywhere"
"WORA"

Java was developed by James Gosling in 1995.

Features of Java :-

- Simple
- Object-oriented
- Platform independent
- Secure
- Robust
- Portable
- Multithreaded

Java file extension $\rightarrow$.java

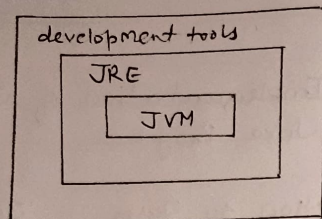
} A byte file is created, that can be used at Any platform.
extension $\Rightarrow$.class

Application of Java :-

- Web Applications
- Android Applications / Desktop Application
- Banking System
- Game development
- API development
- Backend development

$\rightarrow$ It is a case sensitive language.

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JDK

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Introduction to JDK :-

JDK → Java development kit

It is a software package used to develop, compile & run java programs. It includes JVM, JRE & development tools like Javac & java.

JVM → Run

JRE → library

JDK → development tool

working process -

Source Code → javac → Bytecode → JVM → Output
(.java) (.class)

Basic Syntax of Java :-

```
import java.util.*;
```

```
class ClassName {
```

```
main → public static void main(String[] args)
```

```
method {
```

```
$
```

```
// Statements ;
```

```
}
```

```
} Block of code
```

```
}
```

Static keyword allows you to compile without any object. Only 1 main method is allowed

Note

javac Demo → create byte file

java Demo → run the file

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output

Hello from Kanchan!

Output

Hello Java!

I'm Kanchan.

SOP → System (default built in class)
out (object)
print (method or println)

Program 1 Single SOP.

To Print single statement

```
class Demo  
{
```

```
    public static void main(String[] args)  
    {
```

```
        System.out.println("Hello from Kanchan");  
    }
```

```
}
```

Program 2 Double SOP

To print multiple statement

```
class Demo  
{
```

```
    public static void main(String[] args)  
    {
```

```
        System.out.println("Hello Java!");
```

```
        System.out.print("I'm Kanchan.");
```

```
    }
```

```
}
```

Output

This is is a single line comment *

This is multiple line comments in Java.

Welcome to GIT, Jaipur,

The multiline comment is ended.

Program 3 To write a program to demonstrate different type of comments.

```
/**
```

```
 * packageName
```

```
 // documentation comment
```

```
 */
```

class Comments

```
{
```

```
 public static void main(String[] args)
```

```
 {
```

```
 // This is single line comment.
```

```
 System.out.println("This is single line comment");
```

```
 /* This is multiple line comment in Java.
```

```
 welcome to GIT, Jaipur.
```

```
 The multiline comment is ended.
```

```
 */
```

```
 System.out.println("This is multiple line comment  
 in Java");
```

```
 System.out.println("Welcome to GIT, Jaipur,");
```

```
 System.out.print("The multiline comment is ended");
```

```
 }
```

```
}
```

Result:- Thus, the introduction of Java & JDK was studied & programs were successfully executed of SOP & comments.

Viva Questions

Q1 What is Java?

Ans Java is a high level, object-oriented, platform-independent programming language.

Q2 Who developed Java?

Ans Java was developed by James Gosling in 1995.

Q3 What is JDK & JVM?

Ans JDK is Java development kit used to develop & run the Java programs & JVM is Java virtual machine which executes Java bytecode.

Q4 What is main method in Java?

Ans main() is the entry point of a Java program.

Q5 What are comments in Java?

Ans Comments are non-executable statements used to explain code.